

Introduction to causal inference using machine learning methods

Chapter 5: Additional topics

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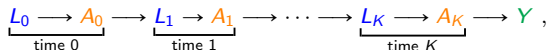
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Motivation for considering time-varying interventions

In many clinical contexts, the treatment of interest is administered in phases over time.

- antihypertensive drug therapy administered daily;
- biphosphonate drug therapy administered weekly;
- injection of antiretroviral suspension administered every month;
- immunosuppressant infusion therapy administered every two months.

The observed data is often of the form



where we have defined components

- L_k = covariates recorded at time k ;
- A_k = treatment assignment at time k ;
- Y = outcome recorded at the end of the study.

Motivation for considering time-varying interventions

We can consider the counterfactual outcome $Y(a_0, a_1, \dots, a_K)$ defined by enforcing treatment assignment $(A_0, A_1, \dots, A_K) = (a_0, a_1, \dots, a_K)$.

This allows to define causal contrasts that address the scientific question of interest.

(Chapters 24-26 of van der Laan & Rose, 2011; Chapter 4 of van der Laan & Rose, 2018; Chapter 19 of Hernán & Robins, 2018)

Weekly alendronate therapy for osteoporosis and one-year incidence of hip fracture:

- L_k = covariates recorded at week k
(e.g., sex, age, dexascan values, thyroid hormone levels, side effects, fracture status);
- A_k = indicator that alendronate was taken at week k ;
- Y = indicator that hip fracture occurred within one year.

We may be interested in the average effect

$$E[Y(1, 1, \dots, 1)] - E[Y(0, 0, \dots, 0)]$$

of year-long weekly alendronate therapy on one-year risk of hip fracture versus no alendronate therapy, or other contrasts defined by values of $(a_0, a_1, \dots, a_{52})$.

Motivation for considering time-varying interventions

Even when the treatment is administered at a single time-point, it is often the case that the data are incompletely recorded in the follow-up period.

- **missing data:** patient did not show up to a scheduled clinic visit;
- **loss to follow-up:** patient moved out of the country and dropped out of the study.

It would be natural then to consider a counterfactual outcome defined by enforcing

- 1 the administration of a particular treatment (baseline only or time-varying);
- 2 complete follow-up and complete recording of data (time-varying).

What would the outcome have been had:

- > the patient taken an experimental treatment regime, the follow-up been complete, and all data been completely recorded?
- > the patient taken a control treatment regime, the follow-up been complete, and all data been completely recorded?

Motivation for considering time-varying interventions

For example, if treatment is only administered at baseline, we could set:

L_k = covariates recorded at time k ;

A_0 = treatment assignment at time 0 (i.e., at baseline);

A_k = indicator that, at time k , patient has not yet been lost to follow-up and all measurements on this patient are complete;

Y = outcome recorded at the end of the study.

We might then be interested in

$$ATE = E[Y(1, 1, 1, \dots, 1)] - E[Y(0, 1, 1, \dots, 1)] .$$

Motivation for considering time-varying interventions

If treatment is administered over time, we could instead set:

L_k = covariates recorded at time k ;

$A_{k,1}$ = indicator that, at time k , patient has not yet been lost to follow-up and all measurements on this patient are complete;

$A_{k,2}$ = indicator of treatment assignment at time k ;

Y = outcome recorded at the end of the study.

and let $Y(\underbrace{(a_{0,1}, a_{0,2})}_{a_0}, \underbrace{(a_{1,1}, a_{1,2})}_{a_1}, \dots, \underbrace{(a_{K,1}, a_{K,2})}_{a_K})$ be the counterfactual defined by

$$(A_0, A_1, \dots, A_K) = (a_0, a_1, \dots, a_K) ,$$

where we write $A_k := (A_{k,1}, A_{k,2})$.

We might then be interested in

$$E[Y((1, 1), (1, 1), \dots, (1, 1))] - E[Y((1, 0), (1, 0), \dots, (1, 0))] .$$

Identification of average treatment effects

Our goal is to contrast the mean outcome under various sequences of interventions occurring over time. We focus on treatment profile $(a_0, a_1, \dots, a_K) = (1, 1, \dots, 1)$ but other profiles can be dealt with similarly.

(Sequentially) randomized trial

We can imagine conducting a trial in which, at each of these time-points, individuals are randomized to one of the possible interventions.

In this case, at each time-point, the intervention assignment is independent of the possible counterfactual outcomes.

$$Y(1, 1, \dots, 1) \perp A_0, \quad Y(1, 1, \dots, 1) \perp A_1, \quad \dots, \quad Y(1, 1, \dots, 1) \perp A_K.$$

Identification of average treatment effects

Our goal is to contrast the mean outcome under various sequences of interventions occurring over time. We focus on treatment profile $(a_0, a_1, \dots, a_K) = (1, 1, \dots, 1)$ but other profiles can be dealt with similarly.

Observational study

In an observation study, there are often factors that influence both the intervention assignment mechanisms and the counterfactual outcome distribution.

Examples of time-varying confounding:

- a patient may discontinue chemotherapy because they have ceased to respond, which may itself be a marker of disease progression;
- a patient may have ceased smoking because they developed respiratory symptoms, which may be a sign of lung cancer.

Identification of average treatment effects

Our goal is to contrast the mean outcome under various sequences of interventions occurring over time. We focus on treatment profile $(a_0, a_1, \dots, a_K) = (1, 1, \dots, 1)$ but other profiles can be dealt with similarly.

Observational study

The vector of time-varying covariates $(L_0, L_1, \dots, L_K)$ can be used to deconfound the relationship between Y and $(A_0, A_1, \dots, A_K)$ provided

$$Y(1, 1, \dots, 1) \perp A_0 \mid L_0, \quad Y(1, 1, \dots, 1) \perp A_1 \mid \bar{L}_1, A_0 = 1, \quad \dots \\ Y(1, 1, \dots, 1) \perp A_K \mid \bar{L}_K, \bar{A}_{K-1} = 1_K,$$

where the symbol 1_j is used to denote a vector $(1, 1, \dots, 1)$ of length j .

In other words, at each time-point, intervention assignment is randomized *within each stratum* defined by recorded patient history up to that point, among patients who have received the intervention of interest so far.

This is referred to as the **sequential randomization** (or **exchangeability**) **condition**.

Identification of average treatment effects

Our goal is to infer what the mean outcome would be in the target population under the multi time-point intervention of interest.

We must be able to **observe the intervention of interest for each different “type” of individual** (as defined by recorded covariates) from this population:

- $P(A_0 = 1 \mid L_0 = \ell_0) > 0$ for each possible ℓ_0 ;
- $P(A_1 = 1 \mid \bar{L}_1 = \bar{\ell}_1, A_0 = 1) > 0$ for each possible $\bar{\ell}_1$;
- ...
- $P(A_K = 1 \mid \bar{L}_K = \bar{\ell}_K, \bar{A}_K = 1_K) > 0$ for each possible $\bar{\ell}_K$.

As before, this is referred to as the **positivity condition**.

Identification of average treatment effects

We first focus on a setting with three time-points ($K = 2$).

$$E[Y \mid A_1 = 1, L_1, A_0 = 1, L_0]$$

$$= E[Y(1, 1) \mid A_1 = 1, L_1, A_0 = 1, L_0] = E[Y(1, 1) \mid L_1, A_0 = 1, L_0]$$

= mean counterfactual outcome among patients treated at time 0, with covariate value L_1 at time 1 and L_0 at time 0

$$E[E[Y \mid A_1 = 1, L_1, A_0 = 1, L_0] \mid A_0 = 1, L_0]$$

$$= E[Y(1, 1) \mid A_0 = 1, L_0] = E[Y(1, 1) \mid L_0]$$

= mean counterfactual outcome among patients with covariate value L_0 at time 0

$$E[E[E[Y \mid A_1 = 1, L_1, A_0 = 1, L_0] \mid A_0 = 1, L_0]]$$

$$= E[Y(1, 1)]$$

= mean counterfactual outcome

Identification of average treatment effects

This can be generalized to an arbitrary number of time-points (i.e., arbitrary K).

Under the sequential randomization and positivity conditions, it can be shown that the mean counterfactual outcome value $E[Y(1, 1, \dots, 1)]$ is given by

$$E\left[E\left[E\left[\dots\left[E\left[E\left(Y \mid \bar{A}_K = 1_{K+1}, \bar{L}_K\right) \mid \bar{A}_{K-1} = 1_K, \bar{L}_{K-1}\right] \dots\right] \mid \bar{A}_1 = 1_2, \bar{L}_1\right] \mid A_0 = 1, L_0\right]\right],$$

where, for any k , we write $\bar{A}_k := (A_0, A_1, \dots, A_k)$ and $\bar{L}_k := (L_0, L_1, \dots, L_k)$.

This is the multi time-point extension of the **G-computation formula** (Robins, 1986).

Identification of average treatment effects

This can be generalized to an arbitrary number of time-points (i.e., arbitrary K).

Under the sequential randomization and positivity conditions, it can be shown that the mean counterfactual outcome value $E[Y(1, 1, \dots, 1)]$ is given by

$$E \left[E \left[E \left[\dots \left[E \left[E(Y \mid \bar{A}_K = 1, \bar{L}_K) \mid \bar{A}_{K-1} = 1, \bar{L}_{K-1} \right] \dots \right] \mid \bar{A}_1 = 1, \bar{L}_1 \right] \mid A_0 = 1, L_0 \right] \right].$$

$\bar{Q}_{K+1}(\bar{L}_K)$
 $\bar{Q}_K(\bar{L}_{K-1})$
 $\bar{Q}_2(\bar{L}_1)$
 $\bar{Q}_1(L_0)$
 $\bar{Q}_0$

Identification of average treatment effects

The idea of inverse probability of treatment weighting naturally suggests a simple identification formula, as in the single time-point setting.

Individuals who received the entire treatment regime of interest are not representative of the target population because of (time-varying) confounding.

What about upweighting their contribution to recover representativeness?

The generalized propensity scores are defined as

$$g_k(\bar{\ell}_k) := P(A_k = 1 \mid \bar{L}_k = \bar{\ell}_k, \bar{A}_{k-1} = 1_k) \text{ for } k = 0, 1, 2, \dots, K.$$

For a patient with partial history $\bar{L}_k = \bar{\ell}_k$, the composite probability of receiving treatment profile $\bar{A}_k = (1, 1, \dots, 1)$ is simply given by

$$\bar{g}_k(\bar{\ell}_k) := \prod_{j=0}^k g_j(\bar{\ell}_j) = g_0(\ell_0)g_1(\bar{\ell}_1)g_2(\bar{\ell}_2) \cdots g_k(\bar{\ell}_k).$$

Identification of average treatment effects

If $\bar{g}_K(\bar{\ell}_K)$ is small, a patient with history $\bar{\ell}_K$ and treatment profile 1_{K+1} is an unlikely occurrence in the sampling population.

This patient needs to serve as stand-in for the many such patients not seen.

The **IPTW identification formula** is given by

$$E[Y(1, 1, \dots, 1)] = E\left[\left\{\frac{A_0 A_1 \dots A_K}{\bar{g}_K(\bar{L}_K)}\right\} Y\right]$$

for treatment profile $(a_0, a_1, \dots, a_K) = (1, 1, \dots, 1)$, and similarly for other profiles.

How does the risk of positivity violations compare to the single time-point setting?

Identification of average treatment effects

The equivalence between the IPTW and G-computation identification formulas can be established through repeated uses of the law of total expectation.

$$\begin{aligned} E \left[\left\{ \frac{A_0 A_1 \dots A_K}{g_0(L_0) g_1(\bar{L}_1) \dots g_K(\bar{L}_K)} \right\} Y \right] &= E \left[E \left[\left\{ \frac{A_0 A_1 \dots A_K}{g_0(L_0) g_1(\bar{L}_1) \dots g_K(\bar{L}_K)} \right\} Y \mid \bar{L}_K, \bar{A}_K \right] \right] \\ &= E \left[\left\{ \frac{A_0 A_1 \dots A_K}{g_0(L_0) g_1(\bar{L}_1) \dots g_K(\bar{L}_K)} \right\} \bar{Q}_{K+1}(\bar{L}_K) \right] \\ &= E \left[\left\{ \frac{A_0 A_1 \dots A_{K-1}}{g_0(L_0) g_1(\bar{L}_1) \dots g_{K-1}(\bar{L}_{K-1})} \right\} \frac{\bar{Q}_{K+1}(\bar{L}_K)}{g_K(\bar{L}_K)} E(A_K \mid \bar{L}_K, \bar{A}_{K-1}) \right] \\ &= E \left[\left\{ \frac{A_0 A_1 \dots A_{K-1}}{g_0(L_0) g_1(\bar{L}_1) \dots g_{K-1}(\bar{L}_{K-1})} \right\} E[\bar{Q}_{K+1}(\bar{L}_K) \mid \bar{L}_{K-1}, \bar{A}_{K-1} = 1_{K-1}] \right] \\ &= E \left[\left\{ \frac{A_0 A_1 \dots A_{K-1}}{g_0(L_0) g_1(\bar{L}_1) \dots g_{K-1}(\bar{L}_{K-1})} \right\} \bar{Q}_K(\bar{L}_{K-1}) \right] = \dots \end{aligned}$$

Failure of naive approach to causal inference

When can causal effects be read off regression models in single time-point settings?

LINEAR MODEL without an interaction between A and W :

$$E(Y \mid A = a, W = w) = \beta_0 + \beta_1 a + \beta_2 w$$

$$E(Y \mid A = 1, W = w) - E(Y \mid A = 0, W = w) = \beta_1$$

$$E[E(Y \mid A = 1, W) - E(Y \mid A = 0, W)] = \beta_1$$

Regression coefficients generally cannot be interpreted as average treatment effects.

Failure of naive approach to causal inference

When can causal effects be read off regression models in single time-point settings?

LINEAR MODEL with an interaction between A and W :

$$E(Y \mid A = a, W = w) = \beta_0 + \beta_1 a + \beta_2 w + \beta_3 aw$$

$$E(Y \mid A = 1, W = w) - E(Y \mid A = 0, W = w) = \beta_1 + \beta_3 w$$

$$E[E(Y \mid A = 1, W) - E(Y \mid A = 0, W)] = \beta_1 + \beta_3 E(W)$$

Regression coefficients generally cannot be interpreted as average treatment effects.

Failure of naive approach to causal inference

When can causal effects be read off regression models in single time-point settings?

GENERALIZED LINEAR MODEL (e.g., logistic model):

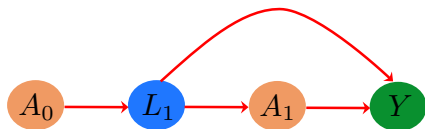
$$E(Y \mid A = a, W = w) = \text{expit}(\beta_0 + \beta_1 a + \beta_2 w)$$

$$E(Y \mid A = 1, W = w) - E(Y \mid A = 0, W = w) = \text{expit}(\beta_0 + \beta_1 + \beta_2 w) - \text{expit}(\beta_0 + \beta_2 w)$$

$$E[E(Y \mid A = 1, W) - E(Y \mid A = 0, W)] = E[\text{expit}(\beta_0 + \beta_1 + \beta_2 W) - \text{expit}(\beta_0 + \beta_2 W)]$$

Regression coefficients generally cannot be interpreted as average treatment effects.

Failure of naive approach to causal inference



$$Y \mid A_1 = a_1, L_1 = \ell_1, A_0 = a_0 \sim \text{Normal}(1 + a_1 + 2\ell_1, 1)$$

$$A_1 \mid L_1 = \ell_1, A_0 = a_0 \sim \text{Bernoulli}(\text{expit}(-1 + \ell_1))$$

$$L_1 \mid A_0 = a_0 \sim \text{Normal}(1 + a_0, 1)$$

$$A_0 \sim \text{Bernoulli}(0.5)$$

Failure of naive approach to causal inference

$$\begin{aligned}Y \mid A_1 = a_1, L_1 = \ell_1, A_0 = a_0 &\sim \text{Normal}(1 + a_1 + 2\ell_1, 1) \\A_1 \mid L_1 = \ell_1, A_0 = a_0 &\sim \text{Bernoulli}(\text{expit}(-1 + \ell_1)) \\L_1 \mid A_0 = a_0 &\sim \text{Normal}(1 + a_0, 1) \\A_0 &\sim \text{Bernoulli}(0.5)\end{aligned}$$

Using the G-computation formula, we can compute mean counterfactual outcomes corresponding to different treatment profiles:

$$\begin{aligned}E[Y(1, 1)] &= E[E[Y \mid A_1 = 1, L_1, A_0 = 1] \mid A_0 = 1] = E[2 + 2L_1 \mid A_0 = 1] = 6 \\E[Y(1, 0)] &= E[E[Y \mid A_1 = 0, L_1, A_0 = 1] \mid A_0 = 1] = E[1 + 2L_1 \mid A_0 = 1] = 5 \\E[Y(0, 1)] &= E[E[Y \mid A_1 = 1, L_1, A_0 = 0] \mid A_0 = 0] = E[2 + 2L_1 \mid A_0 = 0] = 4 \\E[Y(0, 0)] &= E[E[Y \mid A_1 = 0, L_1, A_0 = 0] \mid A_0 = 0] = E[1 + 2L_1 \mid A_0 = 0] = 3\end{aligned}$$

Failure of naive approach to causal inference

Can causal effects be read off the regression of Y on (A_1, L_1, A_0) ?

Effect of differing A_1 values but same A_0 value:

$$E[Y(1, 1) - Y(1, 0)] = 1 = E[Y \mid A_1 = 1, L_1, A_0 = 1] - E[Y \mid A_1 = 0, L_1, A_0 = 1]$$

$$E[Y(0, 1) - Y(0, 0)] = 1 = E[Y \mid A_1 = 1, L_1, A_0 = 0] - E[Y \mid A_1 = 0, L_1, A_0 = 0]$$

Effect of differing A_0 values but same A_1 value:

$$E[Y(1, 1) - Y(0, 1)] = 2 \neq 0 = E[Y \mid A_1 = 1, L_1, A_0 = 1] - E[Y \mid A_1 = 1, L_1, A_0 = 0]$$

$$E[Y(1, 0) - Y(0, 0)] = 2 \neq 0 = E[Y \mid A_1 = 0, L_1, A_0 = 1] - E[Y \mid A_1 = 0, L_1, A_0 = 0]$$

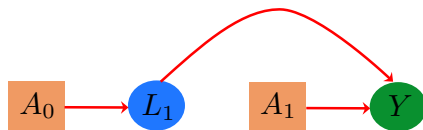
Effect of differing A_1 and A_0 values:

$$E[Y(1, 1) - Y(0, 0)] = 3 \neq 1 = E[Y \mid A_1 = 1, L_1, A_0 = 1] - E[Y \mid A_1 = 0, L_1, A_0 = 0]$$

$$E[Y(1, 0) - Y(0, 1)] = 1 \neq -1 = E[Y \mid A_1 = 0, L_1, A_0 = 1] - E[Y \mid A_1 = 1, L_1, A_0 = 0]$$

Failure of naive approach to causal inference

Why does this happen?



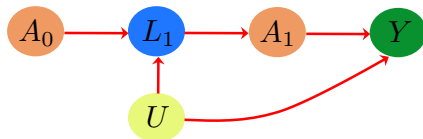
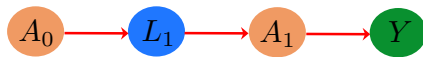
The regression of Y on (A_0, L_1, A_1) fixes L_1 : as such, the causal path between A_0 and Y is blocked. The observed (lack of) association between A_0 and Y is thus not causal.

This is a case of **treatment-confounder feedback**, whose presence often invalidates naive approaches to causal inference (see Chapter 20 of Hernán & Robins, 2018).

Causal methods are even more critical in the context of time-varying interventions!

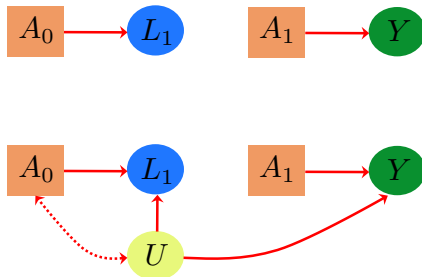
Failure of naive approach to causal inference

Is treatment-confounder feedback present in these examples?



Failure of naive approach to causal inference

Is treatment-confounder feedback present in these examples?



Simple estimation procedures

The IPTW identification formula (slide 13) suggests a simple estimation strategy.

First, we may construct estimates $g_{0n}, g_{1n}, \dots, g_{Kn}$ of propensity scores $g_0, g_1, \dots, g_K$ using our favorite regression estimator for binary outcomes:

- $g_0(\ell_0)$: mean of A_0 given $L_0 = \ell_0$;
- $g_1(\bar{\ell}_1)$: mean of A_1 given $\bar{L}_1 = \bar{\ell}_1$ and $A_0 = 1$;
- ...
- $g_K(\bar{\ell}_K)$: mean of A_K given $\bar{L}_K = \bar{\ell}_K$ and $\bar{A}_{K-1} = 1_K$.

Then, we can compute the corresponding **IPTW estimator** of $E[Y(1, 1, \dots, 1)]$ as

$$\psi_{n, IPTW} := \frac{1}{n} \sum_{i=1}^n \left\{ \frac{A_{0i} A_{1i} \dots A_{Ki}}{g_{0n}(L_{0i}) g_{1n}(\bar{L}_{1i}) \dots g_{Kn}(\bar{L}_{Ki})} \right\} Y_i .$$

Simple estimation procedures

The G-computation formula suggests another natural estimation strategy.

We first illustrate in the three time-point setting, in which case the estimand is simply

$$E \left[E \left[E \left(Y \mid A_1 = 1, L_1, A_0 = 1, L_0 \right) \mid A_0 = 1, L_0 \right] \right] .$$

- 1 Get estimate $\bar{Q}_{2,n}$ of the regression function

$$\bar{Q}_2(\ell_1, \ell_0) := E \left(Y \mid A_1 = 1, L_1 = \ell_1, A_0 = 1, L_0 = \ell_0 \right) ,$$

taking Y as outcome and (A_1, L_1, A_0, L_0) as covariate vector.

- 2 Get estimate F_{1n} of the distribution F_1 of L_1 given $A_0 = 1$ and L_0 , taking L_1 as outcome and (A_0, L_0) as covariate vector, and average $\bar{Q}_{2n}$ over F_{1n} to get

$$\bar{Q}_{1n}(\ell_0) := E_{F_{1n}} \left[\bar{Q}_{2n}(L_1, \ell_0) \mid A_0 = 1, L_0 = \ell_0 \right] = \int \bar{Q}_{2n}(\ell_1, \ell_0) dF_{1n}(\ell_1 \mid \ell_0) .$$

- 3 Average $\bar{Q}_{1n}$ over the empirical distribution F_{0n} of L_0 to get

$$\psi_{G,n} := \bar{Q}_{0n} := E_{F_{0n}} \left[\bar{Q}_{1n}(L_0) \right] = \frac{1}{n} \sum_{i=1}^n \bar{Q}_{1n}(L_{0i}) .$$

Simple estimation procedures

In general, STEP 2 must be performed $K - 1$ times, each time taking the output from the previous step and averaging it out relative to an estimated conditional distribution.

This can be challenging since it requires the user to:

- estimate an entire conditional distribution;
- average out an estimated regression function with respect to this estimate.

Instead, a simpler **sequential regression** approach can be used (Bang & Robins, 2005).

To obtain $\bar{Q}_{1n}$, we could run *our favorite regression method*
on all observations with $A_0 = 1$
using $\bar{Q}_{2n}(L_{1i}, L_{0i})$ as outcome and L_{0i} as covariate.

Simple estimation procedures

Target estimand: $E[E[E(Y | A_1 = 1, L_1, A_0 = 1, L_0) | A_0 = 1, L_0]]$

Y	A_1	L_1	A_0	L_0
0	1	0.7	1	2.1
1	1	-0.2	0	-1.6
1	1	2.0	1	0.3
0	0	6.9	1	1.4
1	0	3.1	0	0.9
0	1	-5.2	1	-3.1
1	0	5.2	1	2.5
0	1	-1.1	1	-0.1

Simple estimation procedures

Target estimand: $E[E(E(Y | A_1 = 1, L_1, A_0 = 1, L_0) | A_0 = 1, L_0)]$

Y	A_1	L_1	A_0	L_0
0	1	0.7	1	2.1
1	1	-0.2	0	-1.6
1	1	2.0	1	0.3
0	0	6.9	1	1.4
1	0	3.1	0	0.9
0	1	-5.2	1	-3.1
1	0	5.2	1	2.5
0	1	-1.1	1	-0.1

STEP 1: Regress Y on L_1 and L_0 among those with $A_1 = A_0 = 1$. $\longrightarrow \bar{Q}_{2n}$

Simple estimation procedures

Target estimand: $E[E(E(Y | A_1 = 1, L_1, A_0 = 1, L_0) | A_0 = 1, L_0)]$

Y	A_1	L_1	A_0	L_0	$\bar{Q}_{2n}(L_1, L_0)$
0	1	0.7	1	2.1	0.34
1	1	-0.2	0	-1.6	0.19
1	1	2.0	1	0.3	0.26
0	0	6.9	1	1.4	0.31
1	0	3.1	0	0.9	0.29
0	1	-5.2	1	-3.1	0.15
1	0	5.2	1	2.5	0.36
0	1	-1.1	1	-0.1	0.25

STEP 1: Regress Y on L_1 and L_0 among those with $A_1 = A_0 = 1$. $\longrightarrow \bar{Q}_{2n}$
Compute $\bar{Q}_{2n}(L_1, L_0)$ for every patient.

Simple estimation procedures

Target estimand: $E[E(E(Y | A_1 = 1, L_1, A_0 = 1, L_0) | A_0 = 1, L_0)]$

Y	A_1	L_1	A_0	L_0	$\bar{Q}_{2n}(L_1, L_0)$
0	1	0.7	1	2.1	0.34
1	1	-0.2	0	-1.6	0.19
1	1	2.0	1	0.3	0.26
0	0	6.9	1	1.4	0.31
1	0	3.1	0	0.9	0.29
0	1	-5.2	1	-3.1	0.15
1	0	5.2	1	2.5	0.36
0	1	-1.1	1	-0.1	0.25

STEP 2: Regress $\bar{Q}_{2n}(L_1, L_0)$ on L_0 among those with $A_0 = 1$. $\rightarrow \bar{Q}_{1n}$

Simple estimation procedures

Target estimand: $E[E(E(Y | A_1 = 1, L_1, A_0 = 1, L_0) | A_0 = 1, L_0)]$

Y	A_1	L_1	A_0	L_0	$\bar{Q}_{2n}(L_1, L_0)$	$\bar{Q}_{1n}(L_0)$
0	1	0.7	1	2.1	0.34	0.40
1	1	-0.2	0	-1.6	0.19	0.22
1	1	2.0	1	0.3	0.26	0.20
0	0	6.9	1	1.4	0.31	0.36
1	0	3.1	0	0.9	0.29	0.21
0	1	-5.2	1	-3.1	0.15	0.34
1	0	5.2	1	2.5	0.36	0.20
0	1	-1.1	1	-0.1	0.25	0.29

STEP 2: Regress $\bar{Q}_{2n}(L_1, L_0)$ on L_0 among those with $A_0 = 1$. $\rightarrow \bar{Q}_{1n}$
Compute $\bar{Q}_{1n}(L_0)$ for every patient.

Simple estimation procedures

Target estimand: $E[E(E(Y | A_1 = 1, L_1, A_0 = 1, L_0) | A_0 = 1, L_0)]$

Y	A_1	L_1	A_0	L_0	$\bar{Q}_{2n}(L_1, L_0)$	$\bar{Q}_{1n}(L_0)$
0	1	0.7	1	2.1	0.34	0.40
1	1	-0.2	0	-1.6	0.19	0.22
1	1	2.0	1	0.3	0.26	0.20
0	0	6.9	1	1.4	0.31	0.36
1	0	3.1	0	0.9	0.29	0.21
0	1	-5.2	1	-3.1	0.15	0.34
1	0	5.2	1	2.5	0.36	0.20
0	1	-1.1	1	-0.1	0.25	0.29

STEP 3: Average out values of $\bar{Q}_{1n}(L_0)$ over all patients to get $\psi_{n,GCOMP} := \bar{Q}_{0n}$.

Improved estimation procedures

Much like in the single time-point case, a hybrid between the G-computation and IPTW estimators can be constructed, and enjoys improved properties.

The **augmented IPTW (AIPTW) estimator** is given by

$$\psi_{n,AIPTW} := \bar{Q}_{0n} + \frac{1}{n} \sum_{i=1}^n \sum_{j=0}^K \left\{ \frac{A_{0i} A_{1i} \dots A_{ji}}{\bar{g}_{jn}(\bar{L}_{ji})} \right\} \left[\bar{Q}_{(j+1)n}(\bar{L}_{(j+1)i}) - \bar{Q}_{jn}(\bar{L}_{ji}) \right],$$

where $\bar{Q}_{0n}$ is simply the G-computation estimator (see slide 23) (Robins ref).

Since it **builds upon estimates of all outcome regressions and propensity scores**, the construction of this estimator requires more effort than for estimators seen so far.

However, this estimator **enjoys double-robustness**, and can be used to construct **valid confidence intervals**, even when flexible learning strategies (e.g., super learner) are used to estimate the outcome regressions and propensity scores.

Improved estimation procedures

What does **double robustness** refer to in the context of multi time-point interventions?

The estimator $\psi_{n,AIPTW}$ built upon estimators

$$\bar{Q}_n := (\bar{Q}_{1n}, \bar{Q}_{2n}, \dots, \bar{Q}_{(K+1)n}) \text{ and } g_n := (g_{0n}, g_{1n}, \dots, g_{Kn})$$

is doubly-robust, in the sense that it is consistent (i.e., hits the target) **provided either $\bar{Q}_n$ hits the target $\bar{Q}_0$ or g_n hits the target g_0 .**

Scenario 1: $\bar{Q}_n$ is consistent

	time 0	time 1	time 2	time 3	...	end
$\bar{Q}$	—	✓	✓	✓	...	✓
g						

Improved estimation procedures

What does **double robustness** refer to in the context of multi time-point interventions?

The estimator $\psi_{n,AIPTW}$ built upon estimators

$$\bar{Q}_n := (\bar{Q}_{1n}, \bar{Q}_{2n}, \dots, \bar{Q}_{(K+1)n}) \text{ and } g_n := (g_{0n}, g_{1n}, \dots, g_{Kn})$$

is doubly-robust, in the sense that it is consistent (i.e., hits the target) **provided either $\bar{Q}_n$ hits the target $\bar{Q}_0$ or g_n hits the target g_0 .**

Scenario 2: g_n is consistent

	time 0	time 1	time 2	time 3	...	end
$\bar{Q}$					...	
g	✓	✓	✓	✓	...	✓

Improved estimation procedures

What does **double robustness** refer to in the context of multi time-point interventions?

The estimator $\psi_{n,AIPTW}$ built upon estimators

$$\bar{Q}_n := (\bar{Q}_{1n}, \bar{Q}_{2n}, \dots, \bar{Q}_{(K+1)n}) \text{ and } g_n := (g_{0n}, g_{1n}, \dots, g_{Kn})$$

is doubly-robust, in the sense that it is consistent (i.e., hits the target) **provided either $\bar{Q}_n$ hits the target $\bar{Q}_0$ or g_n hits the target g_0 .**

Scenario 3: g_n is consistent at early times, $\bar{Q}_n$ is consistent at later times

	time 0	time 1	time 2	time 3	...	end
$\bar{Q}$			✓	✓	...	✓
g	✓	✓				

Improved estimation procedures

The **targeted maximum likelihood estimation (TMLE)** framework provides a recipe for **constructing a G-computation estimator that is also doubly-robust**.

(Bang & Robins 2005; van der Laan & Gruber, 2012)

This estimator is constructed like the 'sequential regression' form of the G-computation estimator, but includes a **refinement step** after each $\bar{Q}$ estimator is obtained.

Using a given estimator g_n of g_0 , the algorithm proceeds as follows:

- (1) build estimate $\bar{Q}_{(K+1)n}$ of $\bar{Q}_{K+1}$ using your favorite regression tool;
- (2) refine $\bar{Q}_{(K+1)n}$, say to $\bar{Q}_{(K+1)n}^*$, to make it a targeted estimate; (★)
- (3) build estimate $\bar{Q}_{Kn}$ of $\bar{Q}_K$ by regressing $\bar{Q}_{(K+1)n}^*(\bar{L}_K)$ using your favorite regression tool;
- (4) refine $\bar{Q}_{Kn}$, say to $\bar{Q}_{Kn}^*$, to make it a targeted estimate; (★)
- ...
- (2K+1) build estimate $\bar{Q}_{1n}$ of $\bar{Q}_1$ by regressing $\bar{Q}_{2n}^*$ using your favorite regression tool;
- (2K+2) refine $\bar{Q}_{1n}$, say to $\bar{Q}_{1n}^*$, to make it a targeted estimate; (★)
- (2K+3) take final estimate to be the G-computation estimator $\psi_{n,TMLE} := \bar{Q}_{0n}^* := \frac{1}{n} \sum_{i=1}^n \bar{Q}_{1n}^*(L_{0i})$.

Improved estimation procedures

What do these so-called **targeting steps** consist of?

STEP (2): Get slope estimate β_n from logistic regression with outcome Y , single covariate

$$Z := (A_0 A_1 \dots A_K) / \bar{g}_{Kn}(\bar{L}_K)$$

and offset term $\text{logit } \bar{Q}_{(K+1)n}(\bar{L}_K)$ using only data with $A_0 = A_1 = \dots = A_K = 1$.

Set $\bar{Q}_{(K+1)n}^* := \text{expit}[\text{logit } \bar{Q}_{(K+1)n} + \beta_n / \bar{g}_{Kn}]$.

STEP (4): Get slope estimate β_n from logistic regression with outcome $\bar{Q}_{(K+1)n}^*$, single covariate

$$Z := (A_0 A_1 \dots A_{K-1}) / \bar{g}_{(K-1)n}(\bar{L}_{K-1})$$

and offset term $\text{logit } \bar{Q}_{Kn}(\bar{L}_{K-1})$ using only data with $A_0 = A_1 = \dots = A_{K-1} = 1$.

Set $\bar{Q}_{Kn}^* := \text{expit}[\text{logit } \bar{Q}_{Kn} + \beta_n / \bar{g}_{(K-1)n}]$.

STEP (2K+2): Get slope estimate β_n from logistic regression with outcome $\bar{Q}_{2n}^*$, single covariate

$$Z := A_0 / \bar{g}_{0n}(\bar{L}_0)$$

and offset term $\text{logit } \bar{Q}_{1n}(\bar{L}_0)$ using only data with $A_0 = 1$. Set $\bar{Q}_{1n}^* := \text{expit}[\text{logit } \bar{Q}_{1n} + \beta_n / \bar{g}_{0n}]$.

Improved estimation procedures

Properties of estimation procedures outlined

	difficulty	$\bar{Q} + \bar{g}$		$\bar{Q} + \bar{g}$		$\bar{Q} + \bar{g}$	
		target	ci	target	ci	target	ci
IPTW	+			✓		✓	
G-COMP	++	✓				✓	
AIPTW	+++	✓		✓		✓	✓
TMLE	++++	✓	✓	✓	✓	✓	✓

$\bar{Q} + \bar{g}$: outcome regressions estimated well but not propensity scores

$\bar{Q} + \bar{g}$: propensity scores estimated well but not outcome regressions

$\bar{Q} + \bar{g}$: outcome regressions and propensity scores estimated well

target : does the estimator hit the right target?

ci : is valid inference possible and readily available, even when flexible learning strategies (such as Super Learner) are used?

Improved estimation procedures

There is substantial work underway to produce novel estimators with even better properties. All of these innovations are based on the idea of TMLE.

■ Enhanced robustness

- Consistent estimation is possible under a wider range of scenarios than depicted on slides 27, 28 and 29.
- For this, more complicated procedures are needed and are being developed. (Luedtke et al. 2017; Rotnitzky et al., 2017)

■ Robust inference

- Typical double-robustness only refers to consistency.
- However, constructing doubly-robust CI and tests is a much more important task. It is also very difficult in multi time-point settings. (Benkeser et al., 2017)

■ Targeted estimation of propensity scores

- Particularly when there are many potential confounders, good performance may be difficult to achieve in smaller samples using the methods described so far.
- Collaborative TMLE allows a smarter, data-driven selection of propensity score estimators to improve performance in such cases. (van der Laan & Gruber, 2010)

Devenir Après Interruption de la FIV (DAIFI) study

(Chapter 25 of van der Laan & Rose, 2011)

- > **Motivating question:** How successful is IVF therapy in France?
- > **Study sample:** All women who received a first IVF cycle at two French IVF units between 1998 and 2002 and were under 42 years of age at initiation.
- > **Intervention considered:** Four successive IVF cycles.
- > **Outcome:** Successful delivery arising from IVF.
- > **Observational challenges:**
 - Some couples abandon mid-course without a successful delivery.
 - Common factors likely influence discontinuation and overall chance of success.
- > **Observed probability of success:**

1st cycle	2nd cycle	3rd cycle	4th cycle
22%	32%	35%	37%

Devenir Après Interruption de la FIV (DAIFI) study

(Chapter 25 of van der Laan & Rose, 2011)

> **Data structure:**

- L_0 = information recorded at baseline (first cycle)
(e.g., age, center, # of embryos transferred or frozen, success of first cycle)
- A_0 = second cycle attempted
- L_1 = success of first two cycles
- A_1 = third cycle attempted
- L_2 = success of first three cycles
- A_2 = fourth cycle attempted
- Y = success of first four cycles

> **Causal estimand of interest:** $E[Y(1, 1, 1)]$

> **Result of TMLE analysis:** 50.5% (95%CI: 48.0–53.0)

HVTN 505 study

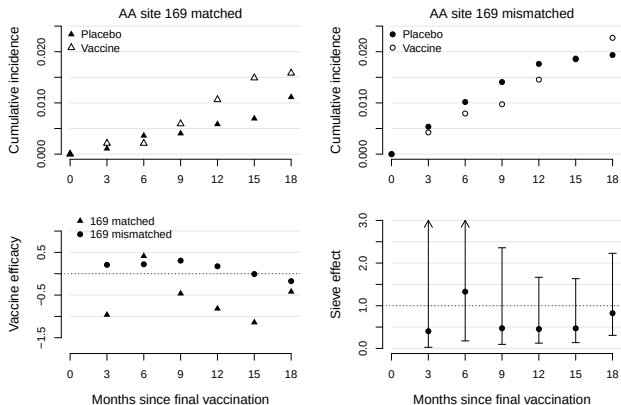
(Chapter 11 of van der Laan & Rose, 2018)

- > **Motivating question:** Does novel HIV vaccine exhibit sieve effects?
- > **Study sample:** Phase II preventive HIV vaccine efficacy trial. 2,405 participants were randomized 1:1 to receive candidate vaccine or placebo.
- > **Interventions considered:** (Active vs control vaccine) + (no loss to follow-up).
- > **Outcome:** Infection with specific genotypes of HIV.
- > **Observational challenges:**
 - Participants may have unblinded using home HIV tests – higher dropout in control arm.
 - Risk behaviors may be informative of participant dropout and HIV infection risk.

Illustrations in practice

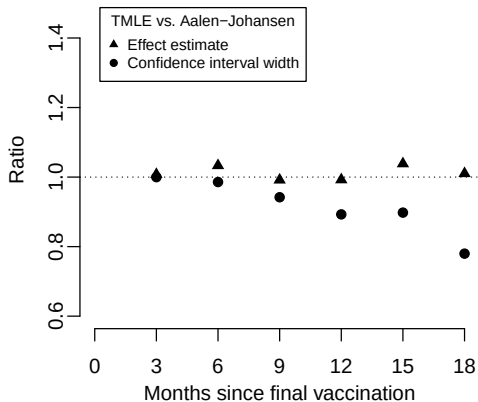
HVTN 505 study

(Chapter 11 of van der Laan & Rose, 2018)



HVTN 505 study

(Chapter 11 of van der Laan & Rose, 2018)



Positivity violations are not uncommon, especially in multi time-point settings.
(Petersen, 2012)

- They may happen **in a study sample by chance** (i.e., practical violations).
 - Can use estimators that borrow more information from other patients.
- They may instead be a **fundamental feature** of the particular combination of population and treatment considered.
 - Can **change the reference population** to exclude subgroups for which the average treatment effect cannot be learned.
 - Can instead **focus on more realistic interventions** for which positivity holds, such as dynamic treatments reflecting clinical practice.

Dynamic treatment rules

Counterfactuals defined by fixed treatment profiles are often neither particularly clinically interesting nor supported by data.

Treatment decisions are usually dynamic and incorporate real-time patient information.

Example: mercaptopurine in IBD patients

- static intervention: 'always treat' versus 'never treat'
- if patient develops signs of liver damage, therapy is usually stopped
- liver function is a time-varying confounder between treatment status and survival
- if poor liver function is a contraindication for therapy, it may not be possible to observe treatment adherence among patients with recent liver failure
- static intervention is unrealistic and not identifiable
- dynamic intervention: 'treat while liver function permits it' versus 'never treat'

$$d(t) = \begin{cases} 1 & : \text{if recent liver function is adequate} \\ 0 & : \text{otherwise} \end{cases}.$$

Dynamic treatment rules

Counterfactuals can be naturally defined in terms of **dynamic treatment rules encoding treatment decisions that possibly depend on current and past patient info.**

In the mercaptopurine example, we may want to learn about the average effect

$$ATE(d, d_0) := E[Y(d)] - E[Y(d_0)]$$

of rule d enforcing treatment whenever liver function permits it and rule d_0 enforcing no mercaptopurine use.

All methods discussed so far can be adapted for use with dynamic treatment rules. Wherever we imposed $\bar{A}_k = \bar{a}_k$, we now instead set $\bar{A}_k$ to equal $d(\bar{L}_k)$ or $d_0(\bar{L}_k)$.

Dynamic treatment rules

A vast subfield of causal inference focuses on the quest for optimal rules.
(Chakraborty & Moodie, 2013)

Of all candidate treatment rules, which one results in the most beneficial average treatment effect?

This is referred to as an **optimal dynamic treatment regime**.

Finding the optimal rule and constructing confidence intervals for the average effect corresponding to this rule using the same data is challenging.

(see, e.g.: Laber et al., 2014; Luedkte & van der Laan, 2016)

Missing outcomes

In observational studies and clinical trials, the outcome Y is typically not observed for every participant.

- People withdraw consent, move out of area, stop showing up for study visits, . . .
- The observed data unit is then $O := (W, A, \Delta, \Delta Y)$, with Δ denoting an indicator of having a measured outcome.

People who are lost may be meaningfully different from those who remain under study.

- What if every patient who had more serious disability three months after imaging did not show up for their one-year visit?
- If the analysis excludes these patients, we are making inference about treatment efficacy *in a possibly very different population*.

Can we account for possibly informative missingness?

Missing outcomes

Ideally, for each patient in the population, we would like to:

- 1 Set treatment to $A = 1$ **and** observe patient at end of study (i.e., set $\Delta = 1$).
- 2 Set treatment to $A = 0$ **and** observe patient at end of study (i.e., set $\Delta = 1$).

The relevant counterfactual means are denoted by $E[Y(1, 1)]$ and $E[Y(0, 1)]$, where $Y(a, \delta)$ is the counterfactual outcome corresponding to setting $A = a$ and $\Delta = \delta$.

The difference $E[Y(1, 1)] - E[Y(1, 0)]$ is the ATE of interest in the full population.

What assumptions do we need to identify these parameters?

Missing outcomes

Previously, we required the randomization condition

$$Y(1) \perp A \mid W$$

which implies the treatment is randomized within strata of recorded covariates.

This will hold if:

- the study guarantees it by design (e.g., stratified randomized trial);
- all potential confounders have been recorded.

We now require the stronger randomization condition

$$Y(a, 1) \perp A \mid W \text{ and } Y(a, 1) \perp \Delta \mid W$$

implying that treatment **and** missingness are randomized within strata.

It is generally difficult to ensure the latter through study design. We must hope that all potential confounders have been recorded.

Missing outcomes

Previously, we required the positivity assumption

$$P(A = a \mid W = w) > 0 \text{ for every plausible value } w$$

which implies all patients may potentially be assigned to treatment group $A = a$.

We now require the stronger positivity condition that, for all plausible value w ,

$$P(A = a \mid W = w) > 0 \text{ and } P(\Delta = 1 \mid A = a, W = w) > 0 .$$

Each patient must have had the chance to receive treatment **and** remain under study.

Missing outcomes

If the **randomization and positivity conditions hold**, then by the G-computation formula

$$\begin{aligned} E[Y(a, 1)] &= E[E(\Delta Y \mid A = a, \Delta = 1, W)] \\ &= \sum_w E(\Delta Y \mid A = a, \Delta = 1, W = w)P(W = w) . \end{aligned}$$

We can define the subgroup-specific average treatment effect as

$$SATE(w) := E(\Delta Y \mid A = 1, \Delta = 1, W = w) - E(\Delta Y \mid A = 0, \Delta = 1, W = w) .$$

G-computation pools subgroup-specific treatment effects across target population:

$$\begin{aligned} ATE &= E[Y(1, 1)] - E[Y(0, 1)] \\ &= \sum_w SATE(w)P(W = w) . \end{aligned}$$

The G-computation excluding missing observations instead gives

$$\sum_w SATE(w)P(W = w \mid \Delta = 1) .$$

Missing outcomes

The **inverse-probability-of-treatment-weighted (IPTW)** identification formula can also be modified to account for missing outcomes.

$$E[Y(a, 1)] = E \left[\frac{I(A = a, \Delta = 1)\Delta Y}{P(A = a, \Delta = 1 | W)} \right] = E \left[\frac{I(A = a, \Delta = 1)\Delta Y}{P(A = a | W)P(\Delta = 1 | A = a, W)} \right]$$

This is a weighted average of the outcome of patients with $A = a$, weighted according to their propensity of being assigned to group $A = a$ **and** remaining under study.

If $P(A = 1 | W = w) = .05$, a patient with $W = w$ had a 5% chance of being treated. Before, this patient stood in for approximately 19 similar patients not treated.

Now, this patient must also stand in patients who had a missing outcome.

If $P(\Delta = 1 | A = 1, W = w) = 0.95$, a treated patient with $W = w$ had a 95% chance of completing the study. This patient has weight

$$\frac{1}{P(A = 1 | W = w)P(\Delta = 1 | A = 1, W = w)} = \frac{1}{0.05 \times 0.95} \approx 21 .$$

Missing outcomes

The AIPTW estimator of ψ_1 can also be modified to account for missing outcomes:

$$\underbrace{\frac{1}{n} \sum_{i=1}^n \left[\frac{I(A_i = 1, \Delta_i = 1)}{g_n(W_i)} \right] Y_i}_{\text{IPTW estimator}} + \underbrace{\frac{1}{n} \sum_{i=1}^n \left[1 - \frac{I(A_i = 1, \Delta_i = 1)}{g_n(W_i)} \right] \bar{Q}_n(1, W_i)}_{\text{augmentation term}} .$$

Here, g_n is an estimator of the extended propensity score

$$g(w) := P(\Delta = 1, A = 1 \mid W = w) ,$$

the conditional probability of being assigned $A = 1$ and remaining under observation.

A modified version of TMLE can be implemented, by fitting the augmentation regression using the covariate

$$Z^1 = \frac{I(A = 1, \Delta = 1)}{g_n(W)} .$$

The estimate g_n could simply be constructed by fitting a regression of the binary outcome $\tilde{A} := I(A = 1, \Delta = 1)$ on covariate vector W .

Missing outcomes

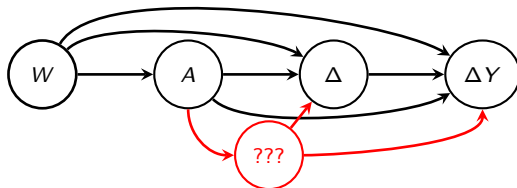
Recall that the randomization condition requires

$$Y(a, 1) \perp A \mid W \text{ and } Y(a, 1) \perp \Delta \mid W$$

implying that treatment **and** missingness are randomized within strata.

This must preclude the possibility that **confounding** happens after A is assigned.

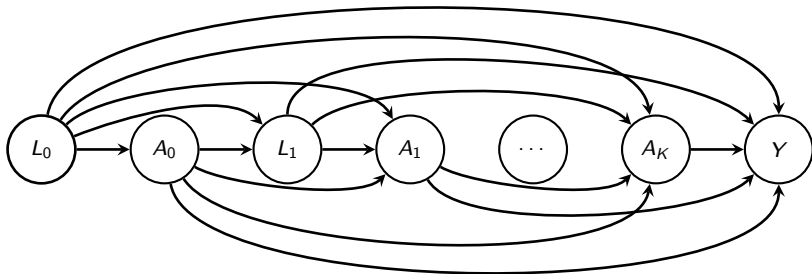
Do we believe that **nothing** happens between treatment assignment and end of study that influences a patient's outcome and probability of remaining under study?



Missing outcomes

We then have time-varying confounding, as is common in longitudinal data.

- $K + 1$ treatment decisions (or missingness indicators);
- Time-varying confounders measured between each treatment decision;
- Counterfactual outcome is $Y(\bar{a})$, where $\bar{a} := \{a_0, a_1, \dots, a_K\}$;
- How do we identify and estimate $E[Y(\bar{a})]$?



Key points: missing outcomes

- Missingness is a problem in both observational data **and** clinical trials.
- Ignoring missing outcome data can bias estimates of causal effects.
- Typically, we can view censoring as another “treatment” decision and, if no time-varying confounding, we can easily modify G-computation, IPTW, and efficient approaches accordingly.
- If time-varying confounding, additional identification results and estimation strategies are necessary.

Missing covariates

In practice, covariates may also be missing for certain observations.

Consider the case of two covariates $W = (W_1, W_2)$, where W_2 is missing for some participants. So, in reality, we observe $W_* = (W_1, \Delta_2, \Delta_2 W_2)$.

Just as with missing outcomes, ignoring data with missing covariates could also cause bias in causal effect estimates.

We recommend this general strategy for handling missing covariates:

- 1 Does randomization holds given W_* ?
 - If so, can proceed with standard approach with adjustment for W_1 , Δ_2 and $\Delta_2 W_2$.
 - See Greenland & Finkle (1995) for limitations of this strategy.
- 2 Otherwise, does randomization hold for Δ_2 given W_1 ?
 - If so, treat missing indicator as intervention node, and use methods for multi time-point interventions. Intervention is on (Δ_2, A) in the sequence

$$W_1 \longrightarrow \Delta_2 \longrightarrow W_2 \longrightarrow A \longrightarrow Y .$$

- 3 Otherwise, assess which randomization assumption is more plausible, and refer to section on missing confounders.

Missing covariates

As an example, say W_2 represents a yes/no item on a health questionnaire to which some patients do not answer. In the observed data, W_2 can take values yes, no or NA.

Let $\Delta_2 = I(W_2 \neq \text{NA})$ and $\Delta_2 W_2 = \Delta_2 I(W_2 = \text{yes})$.

How can we determine whether $Y(a) \perp A \mid W_1, \Delta_2, \Delta_2 W_2$?

- Consider a stratum of patients who refuse to answer the questionnaire and have the same value of W_1 .
- Reasonable to assume that treatment is randomized in this stratum?
- Or are there meaningful differences between people in this stratum who do and do not receive treatment?

If reasonable, then

$$E[Y(a)] = E[E(Y \mid A = a, W_1, \Delta_2, \Delta_2 W_2)]$$

and we can proceed using the standard methodology we have discussed.

Missing covariates

There are several options for estimating the outcome regression and propensity score. Considering the propensity score, how could we estimate $P(A = 1 \mid W_1, \Delta_2, \Delta_2 W_2)$?

The standard regression approach treats missingness as its own category, e.g.,

$$P(A = 1 \mid W_1, \Delta_2, \Delta_2 W_2) = \text{expit}(\alpha_0 + \alpha_1 W_1 + \alpha_2 \Delta_2 + \alpha_3 \Delta_2 W_2) .$$

Another option: we could “impute” missing values of W_2 (e.g., based on W_1) and fit a “full data” method on the imputed data. For example, we could

- 1 fit linear regression of W_2 on W_1 in subset of the data with $\Delta_2 = 1$;
- 2 fill in W_2 with predicted value from regression for observations with $\Delta_2 = 0$;
- 3 fit a random forest with outcome A and predictors W_1 and imputed W_2 .

The choice of imputation technique can now be viewed as a tuning parameter for an algorithm. Let the super learner determine which is best!

Missing covariates

If there is residual confounding due to missingness in W_2 , we can ask whether randomization holds for Δ_2 given W_1 .

Can we explain the missingness in W_2 by the fully measured covariate W_1 ?

Example: We typically have some demographic information on all patients. Is this enough to explain why they did not answer questionnaire?

If so, then we might view Δ_2 as an intervention node and define our counterfactual of interest as $Y(1, a)$, the outcome we would observe if we forced $\Delta_2 = 1$ **and** $A = a$.

In such case, we have that:

- positivity requires positive probability of measuring W_2 for all strata of W_1 ;
- we require *sequential randomization*: Δ_2 randomized given W_1 **and** A randomized given $\Delta_2 = 1$, W_1 and W_2 .
- Each of G-computation, IPTW, AIPTW and TMLE can be implemented in a longitudinal (i.e., multi time-point) framework.

Key points: missing covariates

- Ignoring missing covariate data can bias estimates of causal effects.
- Randomization may hold based on the observed missing data structure; however, this cannot be assessed using our observed data. Prior knowledge is key!
- If randomization fails to hold for missing data but we can predict missingness of covariates using available data, we can use approaches for multi time-point interventions.

Unmeasured confounding

In many observational studies, it is known that some confounders were not measured, and the ATE is therefore not identified by G-computation and IPTW formulas.

- Recall that this cannot generally be assessed just by inspecting the data.
- Be prepared to defend your interpretations against skeptics.

There are several ways to proceed in this situation:

- 1 modify interpretation of the results;
- 2 perform sensitivity analysis;
- 3 estimate bounds on causal effects;
- 4 explore alternative identification results.

Unmeasured confounding

Under appropriate conditions, we stated that the causal effect of interest was given by

$$E[E(Y|A = 1, W) - E(Y|A = 0, W)]$$

and that is what we sought to estimate.

However, we can also simply interpret this estimand as a covariate-adjusted marginal association, leaving aside claims of strict causality.

For example, in our BOLD analysis, we could have interpreted our estimand as

the difference in average disability score among patients with identical baseline covariate values who did versus did not receive early imaging, standardized to the baseline covariate distribution in the BOLD patient population.

A few comments:

- For this interpretation, **no causal assumptions are required at all.**
- This is arguably the closest we can get using the data to the true causal effect.
- Sensitivity analysis ties in scientific knowledge to help determine how close to the true effect we may be.

Unmeasured confounding

Sensitivity analysis is a commonly used term in associative analysis.

- How sensitive is a parametric model to including/removing a covariate, including/removing an interaction, etc?
- How well does this parametric model fit the observed data (e.g., model checking)?

For causal analysis, we take an alternative approach to sensitivity analysis.

- **How much unmeasured confounding would there need to be in order to make an observed association disappear?**

A rich literature exists on this topic (see additional reading). We provide an example from Díaz & van der Laan (2013).

- Chagas disease affects 8 million people in Latin America.
- Long incubation periods of the disease (up to 30 years) make clinical trials to study treatment efficacy prohibitively expensive.
- This also makes observational studies difficult as many participants are lost to follow-up over these long periods of time.
- Analysis combines data from 19 observational studies that did not measure participant-level confounders.

Unmeasured confounding

The observed data unit is $O := (A, \Delta, \Delta Y)$, where Y is a binary cure status and Δ indicates whether a patient's cure status was observed at the end of the study.

The causal parameter of interest is the average treatment effect amongst the treated,

$$ATT = E[Y(1) - Y(0)|A = 1] .$$

However, we know that sicker patients tend to receive treatment and also may drop out of studies more often due to their health. As such, ATT cannot be identified based on the observed data.

Nevertheless, consider estimating

$$ATT_{\text{obs}} = E(\Delta Y|A = 1) - E(\Delta Y|\Delta = 1, A = 0) .$$

Can we bound the difference between ATT and ATT_{obs} by a quantity that we understand scientifically?

Unmeasured confounding

The worst-case scenario for the treatment would be if all participants with missing outcomes who received treatment were not cured.

In this case, $E(\Delta Y|A = 1)$ is a conservative gauge of $E[Y(1)|A = 1]$, in the sense that

$$E(\Delta Y|A = 1) \leq E[Y(1)|A = 1] .$$

This gives us the inequality

$$\begin{aligned} ATT_{\text{obs}} - ATT &= \{E(\Delta Y|A = 1) - E(\Delta Y|\Delta = 1, A = 0)\} \\ &\quad - \{E[Y(1)|A = 1] - E[Y(0)|A = 1]\} \\ &\leq E[Y(0)|A = 1] - E(\Delta Y|\Delta = 1, A = 0) \\ &\leq E[Y(0)|A = 1] . \end{aligned}$$

The difference between ATT_{obs} and ATT is at most the probability of “spontaneous cure” amongst the treated, an interpretable quantity.

Unmeasured confounding

We can now study tests of $H_0 : ATT \leq 0$ by studying $H_0 : ATT_{\text{obs}} \leq \delta$ for different choices of δ , a hypothesized spontaneous cure probability.

We would reject H_0 at level $\alpha = 0.05$ for any value of $\delta \leq 0.44$.

- Because sicker patients receive treatment, it is likely that $E[Y(0)|A = 1] \leq E[Y(0)|A = 0]$.
- $\hat{E}(\Delta Y|\Delta = 1, A = 0) = 0.01$, so it is likely that $E(Y(0)|A = 1) \leq 0.44$.

An interpretable bound allows us to conclude that there is likely at least some causal effect of treatment, even with confounding present.

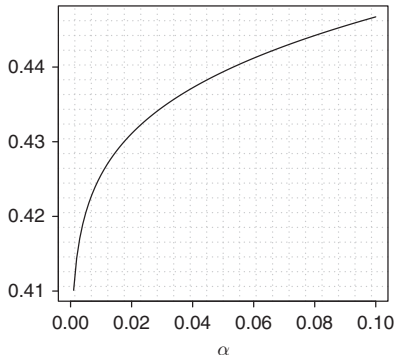


Figure: Choices of δ for which $H_0 : ATT \leq 0$ is rejected at different type I error probabilities (α)

Unmeasured confounding

This form of sensitivity analysis is closely related to approaches in the “partial identifiability” literature (Manski, 1990), wherein bounds on the difference between the causal effect and the observed association are derived.

Usually, very few (if any) assumptions are required to derive bounds, though this sometimes results in highly conservative bounds. For example, we may have generated bounds on the ATE that always include zero.

Unmeasured confounding

In special situations, there may be alternative ways to identify causal effects.

Instrumental variables are a common tool to estimate causal effects with unmeasured confounding (Balke & Pearl, 1997).

- An instrument is a variable Z that is related to A but not to Y .
- Example: Encouragement to seek early imaging.

Under some conditions (Angrist, Imbens & Rubin, 1996), the average treatment effect is identified by

$$ATE = E \left\{ \frac{\overbrace{E(Y|Z=1, W) - E(Y|Z=0, W)}^{\text{conditional effect of } Z \text{ on } Y}}{\underbrace{P(A=1|Z=1, W) - P(A=1|Z=0, W)}_{\text{conditional effect of } Z \text{ on } A}} \right\}.$$

Key points: unmeasured confounding

- If you work with observational data (or even clinical trials data with missing outcomes!), you should worry about unmeasured confounding.
- Ideally, this is corrected at the design stage by collecting a rich set of covariates on trial participants.
- There are several possible approaches to tackle (to the extent possible) unmeasured confounding at the analysis stage.

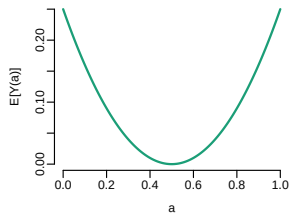
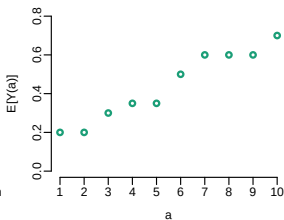
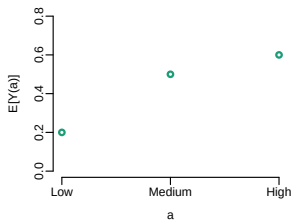
Continuous-valued treatments

In some cases, the treatment of interest A is not binary, but continuous rather.

- Examples: time of early imaging, dose of drug, level of exposure.

We are often interested in a counterfactual outcome under a range of interventions.

- What is the mean outcome if we intervene to set $A = a$ for various $a \in \mathcal{A}$?
- We use $\mathcal{A}$ to denote the set of interventions, which could take only a few values, many values, or an infinite number of values.



Continuous-valued treatments

If $\mathcal{A}$ includes only a few values, all the tools we have developed apply directly.

- We have tools to estimate $E[Y(a)]$ for any particular a .
- Test, e.g., the null hypothesis $H_0 : E[Y(\text{low})] = E[Y(\text{medium})] = E[Y(\text{high})]$.

For statistical reasons beyond the scope of this course, it is not possible to estimate a continuous dose-response curve without making strong assumptions.

- Heuristically, we simply do not observe enough patients with every intervention level we might care about.
- What can we do in this situation?

If $\mathcal{A}$ contains many values, we could still use a multiple degree-of-freedom test to test the null hypothesis that the mean is the same across all $\mathcal{A}$.

- Could we summarize the change in mean across a using a single parameter?

Continuous-valued treatments

We can describe trends using parameters of working models, also known as **marginal structural models**.

- “Working model” implies that we do not necessarily believe this model to be true.
- **Be careful: often, marginal structural models are used in a parametric context in which they are assumed to be true (i.e., not simply working models).**

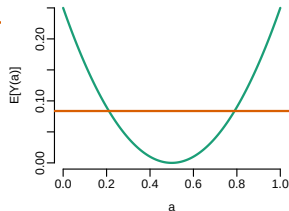
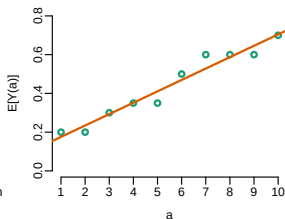
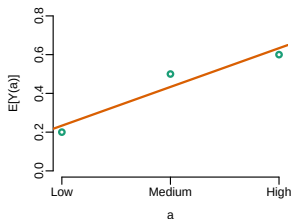
We choose working model $m(a, \beta)$ and define the target parameter as a projection onto this model (Neugebauer, van der Laan, 2007).

- working model: $m(a, \beta) = \beta_0 + \beta_1 a$;
- projection: $\beta_0 = \operatorname{argmin}_{\beta} \sum_{a \in \mathcal{A}} \{E[Y(a)] - m(a, \beta)\}^2$;
- interpretation: the “best-fitting straight line” to the true causal dose-response curve. If $\beta_1 > 0$, then increasing a tends to cause increases in the average outcome.

Continuous-valued treatments

How interesting a given target parameter is depends on the underlying causal curve.

- Linear working models for curves that are monotone may be highly relevant...
- ...but if the curve is parabolic?



Continuous-valued treatments

In order to efficiently estimate parameters of marginal structural models, we need propensity scores $P(A = a \mid W)$ for all $a \in \mathcal{A}$.

- If A has multiple levels, we could use multinomial logistic regression.
- It is also possible to use sequential super learning.

The positivity assumption can be troublesome when treatments have many levels.

- We need positive probability of receiving every treatment in every strata.
- Choosing a weighted projection onto MSM can help alleviate these problems – this is akin to using ‘stabilized weights’:

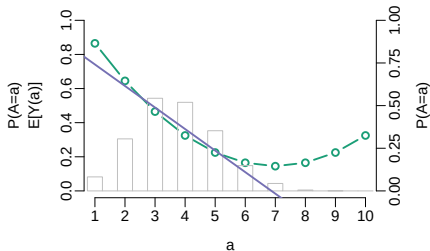
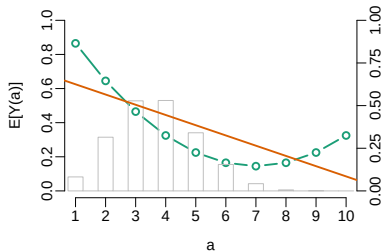
$$\beta_0 = \operatorname{argmin}_{\beta} \sum_{a \in \mathcal{A}} w(a) \{E[Y(a)] - m(a, \beta)\}^2$$

- If $w(a) = P(A = a)$ and a is rare, this level is down-weighted when choosing the best-fitting line.

Continuous-valued treatments

Compare the true value of the parameter for the **unweighted projection** to the **weighted projection**:

- The choice between the two must consider the tradeoff between scientific interest and statistical stability.
- The **unweighted projection** may be more interesting, but we may obtain more precise estimators of the **weighted projection**.



Key points: continuous-valued treatments

- If there are only several discrete categories of treatment, then standard techniques may be applied to each level separately.
- If many levels of ordered treatment, marginal structural models can be useful for summarizing causal dose response curves.
- Positivity issues can be handled by choosing appropriately weighted projections.
- Parameters based on other types of intervention can also be used to study the effect of a continuous treatment on an outcome.

Subgroup effects and interactions

The treatment of interest may have a different effect in different subpopulations.

- Example: Early imaging beneficial in younger patients? In older patients?

Describing whether and how the treatment's efficacy differs across subpopulations is often of interest. For example, what are the policy implications of a treatment with null average effect because:

- it results in worse outcomes in half the population, and better outcomes in half the population, or;
- it has no effect on anyone in the population?

The average treatment effect is not the “wrong” parameter to study, as many people believe. It is still describing the effect of the treatment in the population *on average*.

- G-computation: subgroup-specific treatment effect averaged over subgroups.
- No need for those subgroup-specific effects to be the same!

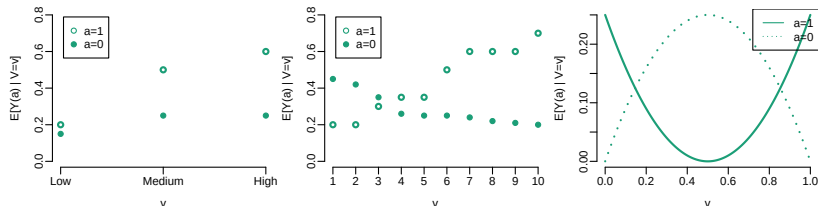
Subgroup effects and interactions

We use $\mathcal{V}$ to denote all the subgroups (defined by baseline covariates) in which we are interested clinically.

- Example: $\mathcal{V} = \{\text{older than 75, younger than 75}\}$
- Example: $\mathcal{V} = \{65\text{yo}, 66\text{yo}, \dots\}$
- Example: $\mathcal{V} = \{65\text{yo men}, 65\text{yo women}, 66\text{yo men}, 66\text{yo women}, \dots\}$

The counterfactual parameters of interest are $E[Y(a)|V=v]$ for $a \in \mathcal{A}$, $v \in \mathcal{V}$.

- For every subgroup v , what is the average outcome under each treatment a .
- We might test whether the average treatment effect is the same for patients across all subgroups.



Subgroup effects and interaction

Identification and estimation of the counterfactual mean in patients with $V = v$ when we assign treatment $A = a$ are straightforward modifications of previous techniques.

G-computation: $E[Y(a)|V = v] = E[E(Y|A = a, V, W)|V = v]$

Averaging is performed relative to distribution of W *only in* stratum $V = v$.

IPTW: $E[Y(a)|V = v] = E\left[\frac{I(A = a, V = v)}{g(v, W)} Y \middle| V = v\right]$

Reconstructing a population of patients with $A = a$ *only in* stratum $V = v$.

Using the observed data, we can estimate

the outcome regression : $\bar{Q}(a, v, w) := E(Y \mid A = a, V = v, W = w)$

the propensity score : $g(v, w) := P(A = a \mid V = v, W = w)$.

Subgroup effects and interactions

In this case, the G-computation estimator of $E[Y(1) \mid V = v]$ is given by

$$\psi_{n,G,1,v} := \frac{1}{\sum_{i=1}^n I(V_i = v)} \sum_{i=1}^n I(V_i = v) \bar{Q}_n(1, v, W_i) .$$

Even though, we can use **all patients** to fit regressions, we are computing the outer empirical mean only using **patients with $V = v$** .

- If there are few of these (e.g., v is a given age), this estimator is unstable.
- If $\mathcal{V}$ assumes an infinite number of values (e.g., v is a given BMI), then we cannot estimate this parameter without making strong assumptions.

We can again make use of nonparametric marginal structural models to summarize these interactions.

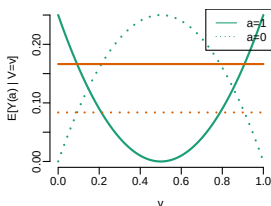
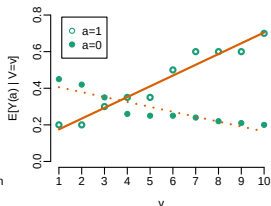
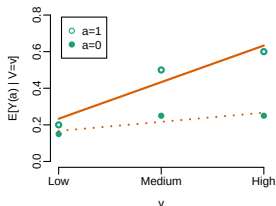
Subgroup effects and interactions

We choose working model $m(a, v, \beta)$ and define the target parameter as a projection onto this model.

- Working model: $m(a, v, \beta) = \beta_0 + \beta_1 a + \beta_2 av$;
- Weighted projection:

$$\beta_0 = \operatorname{argmin}_{\beta} \sum_{a \in \mathcal{A}} \sum_{v \in \mathcal{V}} w(a, v) \{E[Y(a) | V = v] - m(a, v, \beta)\}^2;$$

- Interpretation: The “best-fitting working model” to the true causal dose-response curve across different levels of v . If $\beta_2 > 0$, then there is a trend suggestive of increasing average treatment effect for increasing v .



Key points: subgroup effects and interactions

- Average treatment effect is often relevant even when interactions are present.
- Straightforward extensions of previous results allow us to estimate the average treatment effect within particular strata.
- Comparison of the average treatment effect across these strata constitutes testing for an interaction.
- Marginal structural models may be helpful (often necessary) to estimate interactions between treatment and many subgroups (e.g., defined by a continuous variable).

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